

Problem Statement:

Development of a Real-time Airfare Price Index for India through Automated Web Scrapping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).

Proposed Solution:

- **Automated Airfare Data Collection** – Daily web scraping of airline and travel-aggregator portals to capture fares across major and regional routes.
- **Clean & Reliable Data** – Remove outliers and sold-out flights, separate base fares, taxes and UDF, and generate a data-confidence score.
- **Airfare Price Index (APIx)** – Convert collected fares into a weighted airfare sub-index, using passenger traffic to represent route importance.
- **Price Movement & Monopoly Insights** – Explain fare changes through fuel, demand/seasonality and competition, while flagging unusually high fares on low-competition routes.
- **Dashboard & API for Government** – Provide trend charts, route heatmaps, price explanations, alerts and API access for MoSPI/DGCA/CCI to support CPI-related analysis.





Data Collection (Web Scraping)


Python
Core scripting language


Playwright
Automated, reliable web scraping

Data Processing & Analytics (Cleaning, Anomaly Detection, HHI, SHAP)


pandas
Data cleaning, transformation and preprocessing


scikit-learn
Anomaly detection, HHI (market concentration), ML models


SHAP
Explainable AI (model interpretability)

Database (Storage)


TimescaleDB
Time-series database (optimized for real-time and historical data)

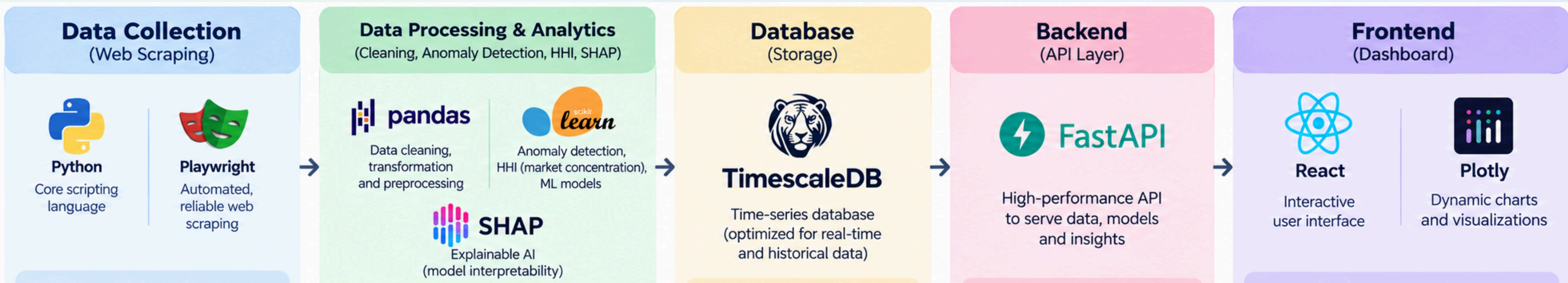
Backend (API Layer)


FastAPI
High-performance API to serve data, models and insights

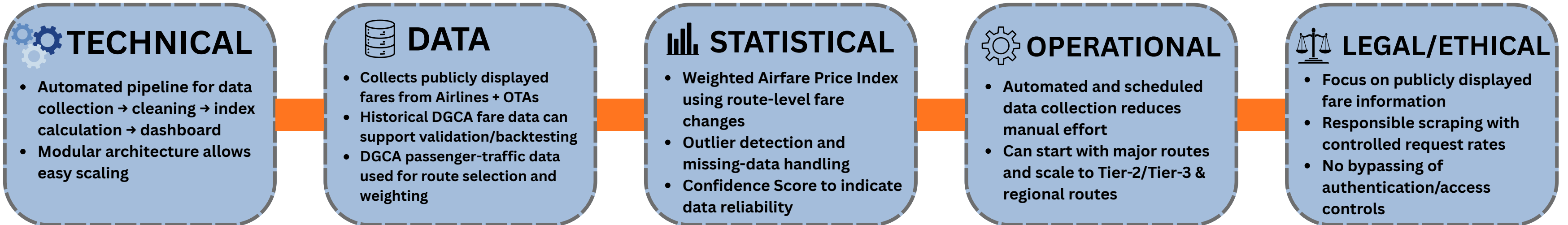
Frontend (Dashboard)


React
Interactive user interface


Plotly
Dynamic charts and visualizations



FEASIBILITY



CHALLENGES & RISKS

- **Dynamic Websites** — Frequent page/API changes
- **Access Restrictions** — CAPTCHA & bot detection
- **Data Quality** — Missing or inconsistent fares
- **Fare Volatility** — Demand & booking-time fluctuations
- **Coverage & Accuracy** — Representative data & index validation

STRATEGIES TO OVERCOME

- **Adaptive Scraping** → Modular scrapers + retries + monitoring
- **Representative Basket** → DGCA traffic-based route selection & weighting
- **Reliability + Backtesting** → Confidence score + Correlation/RMSE/MAPE

1

Improved CPI Accuracy & Timeliness

The proposed APIx provides a continuous, automated airfare price measure that can serve as an improved airfare input for MoSPI's CPI

2

Faster Inflation Monitoring

Instead of waiting for periodic official data, the system generates daily index movements, creating a faster indicator of airfare inflation

IMPACTS

3

Better Regulatory Visibility

The dashboard can provide route-level price trends, heatmaps and surge alerts, helping identify unusual airfare movements

4

Greater Market Transparency

It explains why prices moved, using factors such as fuel/surcharge, demand/seasonality and competition

01

ACCURATE & TIMELY CPI MEASUREMENT

Better Airfare Component for CPI
Provides high-frequency airfare data for a more accurate CPI airfare component.

Real-Time Price Monitoring
Captures daily changes in dynamic airfares.

Improved Inflation Measurement
Provides a timely airfare sub-index to strengthen overall inflation measurement.

03

DATA INDEPENDENCE & RELIABILITY

Reduced Dependence on Airlines
Collects publicly available fare data independently without relying on airline APIs or voluntary data sharing.

Confidence Score
Shows the percentage of expected data successfully collected and highlights data gaps.

Auditable Data Pipeline
Maintains structured records and a reproducible process from data collection to index calculation.

02

COST & RESOURCE EFFICIENCY

Reduces Manual Data Collection
Automates airfare collection from airline and OTA websites.

Scalable Data Collection
Covers multiple routes, airlines and booking windows without significant additional manpower.

Lower Operational Effort
Automates data cleaning, structuring and preparation for index calculation.

04

EVIDENCE-BASED POLICY & REGULATION

Explainable Price Changes
Shows possible reasons for fare changes, including fuel, demand, seasonality and competition.

Unusual Pricing Detection
Flags unusually high fares on routes with limited competition for regulatory review.

Supports Government Decision-Making
Provides API-based data for MoSPI, RBI and CCI/DGCA, supporting inflation analysis, monitoring and policy decisions.

Benefits of the Solution for Government

Real-Time Airfare Price Index for a More Informed India

References:

[1] Outlook Business Desk. (2025, March 24). Indian Airlines vs DGCA: Air India, IndiGo, Others Refuse to Share Airfare Data; Here's Why. Outlook Business.

[Article link](#)

[2] T. Wang, S. Pouyanfar, H. Tian, Y. Tao, M. Alonso Jr., S. Luis, and S.-C. Chen, “A Framework for Airfare Price Prediction: A Machine Learning Approach,” 2019.

[3] F. Polidoro, R. Giannini, R. Lo Conte, S. Mosca, and F. Rossetti, “Web scraping techniques to collect data on consumer electronics and airfares for Italian HICP compilation,” Statistical Journal of the IAOS, vol. 31, pp. 165–176, 2015.

[4] Korkmaz, H. (2024). “Prediction of Airline Ticket Price Using Machine Learning Method.” Journal of Transportation and Logistics.